Lab 3 SE

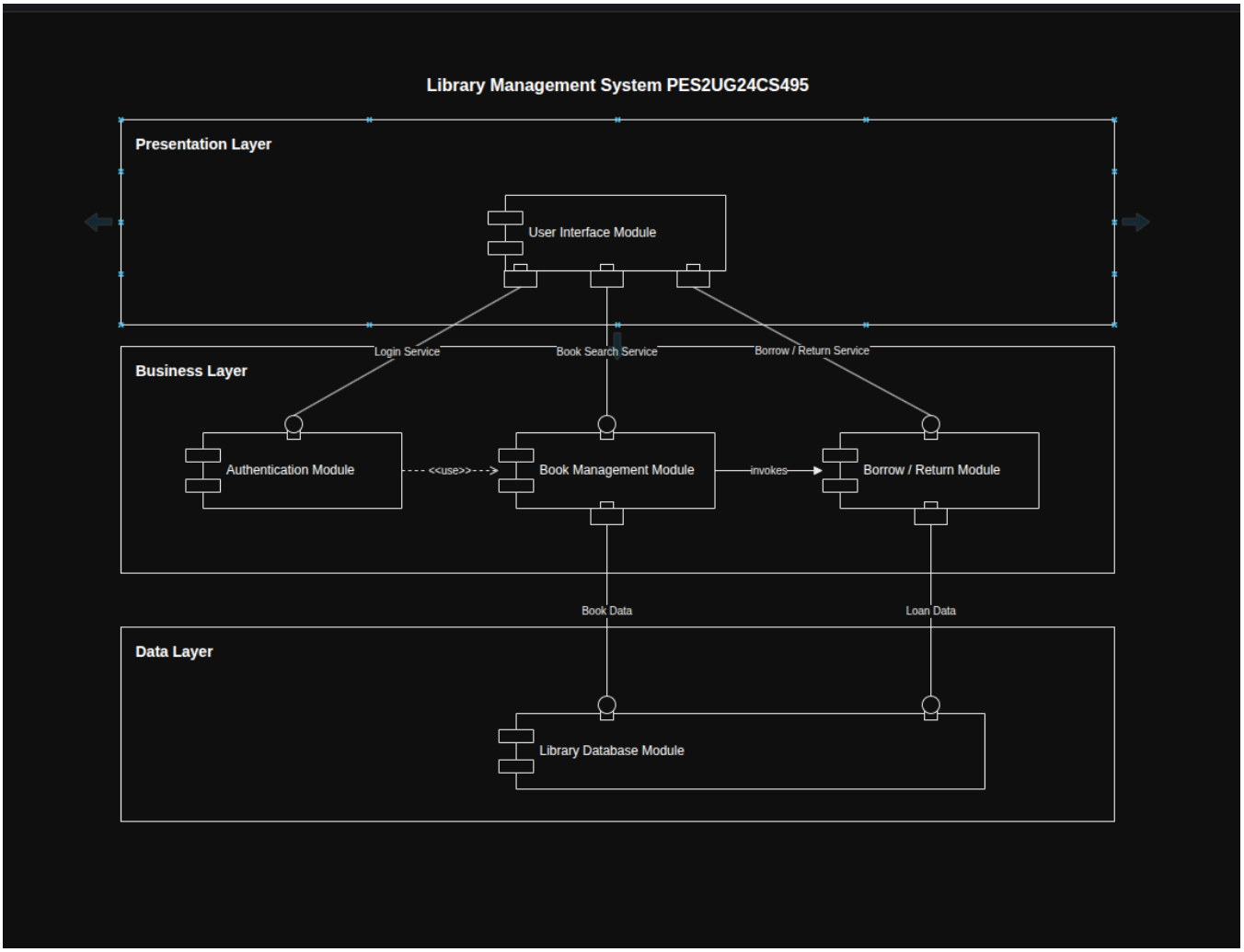
Library Management System – Architectural Justification

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COMPONENT DIAGRAM



Architecture Selection

We chose Layered Architecture for the Library Management System. The system is divided into three layers: Presentation Layer, Business Layer, and Data Layer. This follows the layered architecture approach, which separates presentation, business logic, and data responsibilities.

Architectural Choice

The Presentation Layer contains the User Interface Module, which provides the interface through which users interact with the library system. The Business Layer contains the Authentication, Book Management, and Borrow/Return modules, which handle the core operations of the system. The Data Layer contains the Library Database Module, responsible for storing and retrieving library information.

Reason 1 – Separation of Responsibilities

Layered architecture provides a clear separation between different system responsibilities. User interaction is handled by the User Interface Module, while authentication, book management, and borrowing/returning are handled independently in the Business Layer. Database operations are isolated in the Data Layer. This makes the system easier to understand, maintain, test, and modify.

Reason 2 – Maintainability and Modularity

The system is divided into independent modules that communicate through defined interfaces. For example, the Book Management Module communicates with the Library Database through the Book Data interface, while the Borrow/Return Module uses the Loan Data interface. This modular structure allows individual components to be modified or replaced with minimal impact on other parts of the system.

Security Advantage

The Authentication Module provides a dedicated point for verifying users before they access library operations. Separating authentication from the other business modules helps prevent unauthorized users from directly accessing protected functionality. The <<use>> dependency between Authentication and Book Management represents the dependency on authentication services.

Performance Benefit

The Data Layer centralizes database access, allowing the Book Management and Borrow/Return modules to retrieve only the information required for their operations. Separating database operations from the user interface reduces unnecessary interaction between the presentation and data layers and provides a structured way to manage system requests.

Conclusion

The Layered Architecture provides a simple and organized structure for the Library Management System. The use of components, ports, provided interfaces, required interfaces, assembly connections, and <<use>> dependencies clearly represents how the modules interact while maintaining separation between presentation, business logic, and data storage.